

Jack Norris

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PROFESSIONAL SUMMARY

Software engineer with experience in full-stack web development, applied ML research, and mission-driven technical work across startup, enterprise, and research settings. Background includes React/Next.js, SQL, Python-based ML workflows, and published research, with current work applying AI to biological data.

TECHNICAL PROFICIENCIES

- **Languages (by proficiency):** Python, Typescript/Javascript, C++, SQL, Java, Bash, P4, Kotlin, Swift
- **Frameworks & Libraries:** React, Next.js, Express, Angular, Spring Boot, PyTorch, scikit-learn, pandas, RDKit, Tailwind
- **Tools & Platforms:** Firebase, Git, Docker/Kubernetes, CI/CD pipelines
- **Misc:** Arduino, ESP32, Raspberry Pi, breadboarding & soldering

EDUCATION & HONORS

College of Engineering, University of Arkansas (UARK)

Summa Cum Laude (3.97 GPA)

Major: Honors Computer Science

Graduated: May 2024

Minors: Mathematics & Finance

PROFESSIONAL EXPERIENCE

Peace Corps Paraguay

Caazapá, Paraguay

Community Health Volunteer

May 2024 – Present

- Design and teach community health activities in partnership with local schools and health clinic
- Led coordination of a 5,000lb international book donation shipment supporting 3 rural community libraries
- Co-built and sourced materials for a community outdoor gym and school computer lab

Compound Foundry

Fayetteville, AR

Junior Software Engineer

September 2023 – March 2024

- Built a Shopify extension that automated order printing and shipping workflows for artists, reducing manual fulfillment work and streamlining order processing time
- Technologies used: React/Next.js, Tailwind, Prisma/SQL, GraphQL, Docker, ngrok

Amazon

Seattle, WA & Los Angeles, CA

Software Development Engineer Intern

Summer 2021 & 2022

- Built a no-code interface that enabled product managers to modify JSON-based configurations without direct engineering support, reducing turnaround for internal config changes from 3 weeks to 1 week
- Built experimental SharePlay functionality into Amazon Music, allowing for shared listening of podcasts and music with latency of < 500ms on up to 6 devices
- Technologies used: Swift, React, Scala

J.B. Hunt Transport Services

Fayetteville, AR

Application Development Intern

October 2019 – May 2022

- Shipped production features across 3 applications (carrier management platform, internal interview feedback tool, driver mobile app) with unit tests supporting 80% code coverage
- Technologies used: Angular, Spring Boot, PostgreSQL, Docker/Kubernetes

RESEARCH EXPERIENCE

FIUNA ML/AI Drug Discovery Lab

Asunción, Paraguay

Research Fellow under Dr. Diego Galeano

May 2025 – Present

- Build and evaluate Python-based ML pipelines for chemical compound bioactivity prediction using graph-based and learned molecular representations
- Technologies used: PyTorch, chemprop, ChemBerta, RDKit, scikit-learn, XGBoost, pandas, chemical fingerprinting (ECFP4, FCFP4, MACCS, etc.)

UARK CSCE Computer Networks Lab

Undergraduate Researcher under Dr. Kevin Jin

Fayetteville, AR

May 2022 – May 2024

- Developed P4-based methods for recovering missing phasor measurement unit data in programmable networks for power grid resilience
- Co-authored IEEE SmartGridComm paper on in-network data recovery for PMUs
- Technologies used: P4, Python, Network Socket Programming

PEER-REVIEWED CONFERENCE PUBLICATIONS

- J. Norris, Z. He, Y. Qu, G. Chen, C. Hertzog, D. Jin. “An In-Network Approach for PMU Missing Data Recovery with Data Plane Programmability”, *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, October 2025

PROFESSIONAL DEVELOPMENT & LEADERSHIP

Association for Computing Machinery at UARK

President, former Vice President

Fayetteville, AR

May 2022 – May 2024

- Organized campus networking opportunities, hackathons, internship searches, and computer science outreach for university
- Orchestrated “HogHacks”, the largest hackathon in the Northwest Arkansas area, garnering \$13k in funding, ~200 participants, and involvement with Major League Hacking. Co-built hackathon judging website

McMillon Innovation Studio

Technical Consultant, Senior Project Lead

Fayetteville, AR

August 2020 – May 2024

- Built proof of concept apps and websites for student business ideas in UARK’s student innovation lab
- Co-led an e-paper startup that won \$20k in the Arkansas Governor’s Cup business plan competition
- Led projects for stakeholders at Walmart, UARK, Phigenics, Ox, and the Northwest Arkansas Food Bank

First-Year Engineering Program

Peer Mentor

Fayetteville, AR

August 2021 – May 2024

- Met weekly during the school year with 7+ freshman engineers to offer guidance, advice, and connections as they navigated freshman year

SELECTED PROJECTS

Personal Website & Blog

- Built a responsive React website to showcase projects and host a blog chronicling personal project development and my Peace Corps service
- Developed and deployed a Dockerized Express backend to handle blog email notifications, Spotify API integration, and future database migrations
- Researched and built custom simulations for Conway’s Game of Life, Boids, and the three-body problem
- Technologies used: React, Express, Docker, Tailwind, Firebase Auth & Realtime Database, CI/CD GitHub Actions, Cloudflare DNS, GitHub Pages

Lightweight Automatic Differentiation Engine

- Built and documented the process of building a lightweight automatic differentiation engine in Python to better understand backpropagation and the inner workings of tensors within the PyTorch library
- Utilized the *Tensor* class I developed to train and evaluate a basic neural network on the MNIST dataset, achieving 87.5% classification accuracy
- Technologies used: Jupyter, Python, Numpy

Android Swiss Army Knife App

- Co-built an Android app utilizing the MVVM paradigm that combined some of the most used utility apps into one (level, compass, metal detector, barometer, etc.)
- Technologies used: Android Studio with Kotlin, MVVM, Android Sensor API